

Response to reviewers for

Manuscript Title:

Analysis of human visual experience data

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We thank the editor and the reviewer for their insights and feedback. We have made several changes to the manuscript based on this feedback. We reference these changes point by point, together with the reviewer's remarks. Reviewers comments are written verbatim, with our point by point replies in **bold**.

Reviewer #1:

My outstanding issue with the original manuscript was that no behavioural data was given about the participants whose data was analysed, so the reader had no insight over whether LightLogR was reporting that behaviour correctly.

In the latest revision, the authors have acknowledged that the original data set was atypical (data collected during COVID) and undertook a new analysis of new data from a different source (Bhandari et al, 2022). Given the number of changes to the manuscript, this was clearly a large job and I appreciate the effort the authors expended to try to address my comments.

Unfortunately, I feel the changes only superficially address my original issue. It is still difficult for the reader to understand the behaviour of the participants because this data was also collected during COVID times. According to the abstract (I don't have access to the full, paywalled, article) the participants were ≈ 14 year old school children in the US. Since the details of "lock down" varied greatly between countries, the reader will not know if these children were following normal school hours, whether they were allowed outside during recesses, etc. So again: how is the reader supposed to assess if the LightLogR output looks correct or not?

The authors, in their reply, note that the new data set "has more common values for illuminance" which is reassuring, but a similar statement needs to appear in the manuscript to similarly assure readers. I believe, then, it would considerably help the reader if the authors would modify their sentence introducing the data set by adding some simple details. For example:

"...dataset we will be using comes from a data collection during COVID-19 (14-year-

old school children in Southern US, November 2020, who followed a normal school day, but were not allowed outside during recesses)."

The reader would then have an idea of the illumination intensity and duration of both outdoor (Winter, southerly latitude (I think, based on line 285)) and indoor (typical classroom, constant exposure from $\approx 9\text{am}$ to $\approx 3\text{pm}$) that the participants were exposed to. Again, I don't have access to the article's full text, so the above details will need to be changed accordingly.

The same fix would also be necessary for the second data set. The authors added to the manuscript "For the VEET device, the dataset comes from a pilot data collection of a researcher under naturalistic conditions." but the term "naturalistic" is too nebulous. Was this researcher working in a window-less lab, or an office with a south-facing window? Was this during winter or summer? How many hours, on average, were they at work during the measurement period? Again, a short sentence which replaces the word "naturalistic" with some actual detail would be far more helpful to the reader and satisfy this reviewer.

I'm satisfied with the other changes that I requested in my previous review (fixing column headings, etc).

Our reply:

We thank Reviewer 1 for the constructive feedback. While making changes to address the reviewer's comments we noticed an error in our source for the data. While the source was correct for the old dataset, the new dataset we introduced in the last revision comes from a different source. We apologize for this mistake and have corrected it in the current version. The "Devices" section now reads:

"[...] For the Clouclip device, the dataset was obtained in November 2020 from a 30-year-old male graduate student in Texas, United States, and was previously presented as part of a published study (Bhandari et al., 2021). The light exposure profile reflects relatively limited time outdoors, which is consistent with patterns reported in individuals engaged in intensive academic training and the pandemic during 2020."

"[...] For the VEET device, the dataset comes from pilot data collection of a researcher in Texas, United States, under naturalistic conditions in June 2024 (light and distance) and June 2025 (spectral data). The researcher was working in a typical office building with limited daylight availability during office hours and limited time outdoors overall."

Additionally, we have added a sentence below these two devices that ensures readers that values are typical:

“The distribution of light exposure in both datasets is typical for indoor and partial outdoor times and is discussed further at the beginning of the Results section for light exposure.”

Editor:

Considering the impact on the results (as happened in the last revision), I do agree with the reviewer that these details have to be added, and personally also would like to see a few sentences in the discussion about the need for annotation / qualitative data interpreting this type of data.

Our reply:

We have added a paragraph to the Discussion section to address this aspect. The discussion now reads:

“[...]

As with any behavior-dependent data source, light and distance measurements show substantial interindividual variability, which can lead to marked differences at the individual level. However, the overall processing pipeline generally remains unchanged when applied to group-level analyses. Metric parameters may still require adjustment, particularly when they are determined by device-specific sensor constraints rather than theoretical considerations. For example, the operational definition of near work may vary across populations and devices.

Another important consideration is the role of contextual data in disentangling light and distance patterns into their contributing factors beyond time of day alone. This often requires data sources in addition to the wearable device itself, such as logs or diaries. Zauner and colleagues (2026) describe a strategy for collecting multiple auxiliary data modalities, including activity, sleep, wear time, and other relevant measures. LightLogR provides functions to annotate wearable datasets with such contextual information, enabling more detailed analyses of visual experience data in relation to behaviour and environmental context. A free interactive online training course for the package highlights these advanced analytical pipelines and provides specific examples (Zauner and Spitschan, 2025).

[...]”

We have made another change in the manuscript due to feedback from a reader. In R2, lines 600-601 read: “To use these distance data in the framework shown above for the Clouclip device, a sensible method to condense the data has to be applied. “.

It now reads:

“This spatial resolution necessitates a change in the data format. When we try to apply the preprocessed VEET data to the functions described above for the Clouclip device, we run into issues. This is because each time point is repeated 64 times to include as many spatial positions. LightLogR expects unique time points for all observations within each group (in our case days within participants). To use these distance data in the framework shown above for the Clouclip device, a sensible method to condense the data has to be applied.”